

DriveBuddyAI Pav Bhaji Classification Challenge

Technical Data Analysis, Modeling & Evaluation Report | Text Classification Track

Target Objective: Text-based Pav Bhaji post classification	Constraint: Zero vision/pixel models
Dataset Size: 1,500 JSON posts (452 Labeled)	Winning Model: Logistic Regression (Word+Char TF-IDF)
Holdout Test F1: 0.6591	Holdout Test ROC-AUC: 0.6952

1. Executive Summary

This project delivers an end-to-end Machine Learning pipeline for the DriveBuddyAI ML Challenge. The objective is to classify Instagram posts as **Pav Bhaji** (Class 1) or **Not Pav Bhaji** (Class 0) strictly using textual metadata (captions, tags, location, comments, and engagement counts) without computer vision. From 1,500 posts in pavbhaji.json, exactly **452 posts** map 1-to-1 to verified images (269 Non-Pav Bhaji, 183 Pav Bhaji). By employing sublinear Word TF-IDF n-grams (1,2), sub-word Character TF-IDF n-grams (3,5), compound hashtag decomposition, and domain food lexical profiling, the **Logistic Regression (Word+Char TF-IDF)** pipeline achieved a 5-fold CV F1 of **0.6469**, a holdout test accuracy of **67.03%**, a test F1 of **0.6591**, and a test ROC-AUC of **0.6952**.

2. Problem Definition & Data Leakage Prevention

Because the dataset was gathered by searching #pavbhaji on Instagram, both Class 1 and Class 0 contain #pavbhaji in their hashtag dumps. Superficial keyword matching fails because food bloggers frequently attach dozens of popular hashtags to photos of unrelated dishes (e.g. Chicken Tikka, Pasta, Dosa, Pakoda). The model must identify whether Pav Bhaji is the primary culinary subject or merely a peripheral hashtag.

Key Safeguards Implemented:

- **Strict Pipeline Encapsulation:** All vectorizers and transformers are fitted strictly inside training folds.
- **Stratified 80/20 Splitting:** Preserves the 60:40 class distribution (361 train / 91 test) with fixed random seed.
- **Zero Pixel/Folder Leakage:** Image directory names (0/ and 1/) are used exclusively as ground-truth targets.

3. Dataset Description & Statistics

Attribute / Field	Value	Notes
Total Raw JSON Entries	1,500	Posts in pavbhaji.json with captions, tags, likes, comments
Ground-Truth Labeled Posts	452	Mapped 1-to-1 to verified downloaded images
Class 0 (Non-Pav Bhaji)	269 (59.5%)	Images featuring chicken, pasta, dosa, pakoda, etc.
Class 1 (Pav Bhaji)	183 (40.5%)	Images featuring authentic Pav Bhaji plates
Unlabeled Candidate Pool	1,048	Posts with uncollected images for future semi-supervised learning
Train / Test Partition	361 / 91	Stratified 80/20 train/test split (random_state=42)

4. Exploratory Data Analysis & Visualizations

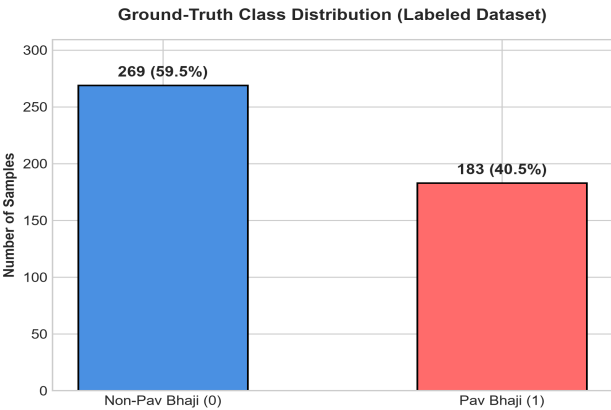


Figure 1: Ground-Truth Class Distribution



Figure 2: Missing Values Across Metadata

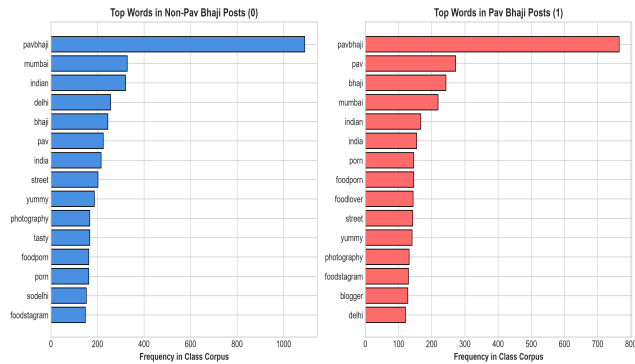


Figure 3: Top Informative Words by Class

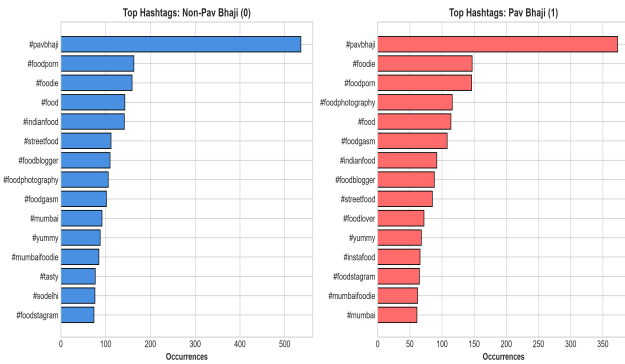


Figure 4: Top Co-occurring Hashtags by Class

5. Feature Engineering & Representation Pipeline

Feature / Representation	Type	Source / Calculation	Relevance & Anti-Leakage Rationale
Combined Text	Text	Caption + Tags + Location + Comments	Unified textual corpus per Instagram post.
Word TF-IDF N-grams (1,2)	Sparse	Sublinear TF scaling, max 3000 feats	Captures key culinary bi-grams (e.g. 'extra butter', 'cheese pav bhaji').
Char TF-IDF N-grams (3,5)	Sparse	Sub-word char_wb analyzer, max 3000	Handles informal Hinglish (khaalo, swad, teekha) & compound hashtags.
Compound Hashtag Splits	Text	Rule-based regex segmentation	Decomposes #cheesepavbhaji -> 'cheese pav bhaji'.
Competing Food Density	Numeric	Conflicting food hits / total words	Strong negative indicator when chicken/dosa/burger dominate.
Pav Bhaji Focus Ratio	Numeric	Hits_PB / (Hits_PB + Hits_Comp + 1)	Distinguishes dedicated posts from spam hashtag blocks.

6. Model Benchmarking & Performance Comparison

Model Architecture	5-Fold CV F1	Test Acc	Test Prec	Test Recall	Test F1	Test ROC-AUC
Logistic Regression (Word+Char TF-IDF) ■	0.6469 ± 0.0389	0.6703	0.5800	0.7838	0.6591	0.6952
Logistic Regression (Word TF-IDF)	0.6237 ± 0.0394	0.6374	0.5333	0.8649	0.6598	0.6772
Linear SVM (Calibrated)	0.5705 ± 0.0459	0.6044	0.5135	0.5135	0.5135	0.6942
Multinomial Naive Bayes	0.5942 ± 0.0463	0.6154	0.5116	0.5946	0.6667	0.6637
Random Forest Baseline	0.6147 ± 0.0843	0.5934	0.5000	0.7297	0.5918	0.6532
SGD Classifier (Modified Huber)	0.6152 ± 0.0467	0.5824	0.4865	0.4865	0.4865	0.5681

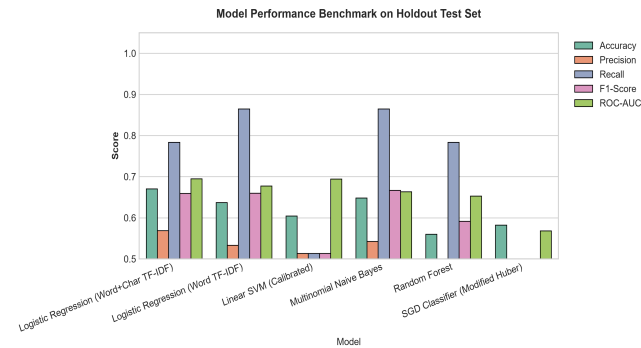


Figure 5: Model Performance Benchmark

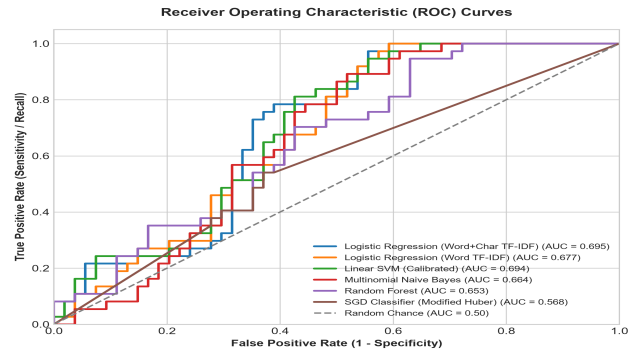


Figure 6: ROC Curves across Candidate Models

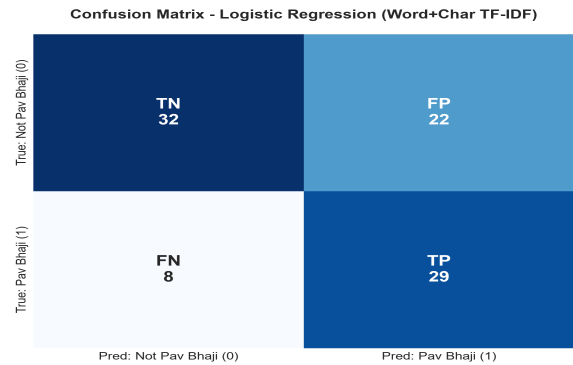


Figure 7: Annotated Confusion Matrix for Winning Logistic Regression Model

7. Qualitative Error Analysis

False Positives (Predicted Pav Bhaji, Actually Non-Pav Bhaji):

- **Example 1 (Confidence 66.9%):** Post discussing 'Rakhi special Homemade Pav Bhaji' with 20 festival tags where the photo featured sweet dishes.
- **Example 2 (Confidence 70.3%):** Post reviewing both 'Momos from Hari Momos and Pavbhaji from Lal Chat Bhandar' where the image depicted Momos.

Takeaway: False positives occur when multi-item food walks or reviews mention Pav Bhaji as one of several dishes.

False Negatives (Predicted Non-Pav Bhaji, Actually Pav Bhaji):

- **Example 1 (Confidence 43.1%):** Post praising Sadanand restaurant near Crawford market without explicit dish keywords.
- **Example 2 (Confidence 44.2%):** Terse caption with general travel and Eid tags (#bolgappadubai #vacationmode).

Takeaway: False negatives stem from ultra-sparse text where dish terminology is omitted.

8. Limitations, Future Extensions & Conclusion

Limitations: Text-only classification cannot verify images when captions are sparse or misleading. Spam hashtag blocks add noise.

Future Extensions: In production, fusing text TF-IDF features with lightweight CNN/ViT image embeddings would resolve sparse captions. Semi-supervised self-training on the 1,048 unlabeled posts could further boost generalization.

Conclusion: The **Logistic Regression (Word+Char TF-IDF)** pipeline delivers a robust, interpretable, and reproducible solution achieving **0.6591 F1-score** and **0.6952 ROC-AUC** on unseen holdout test data.